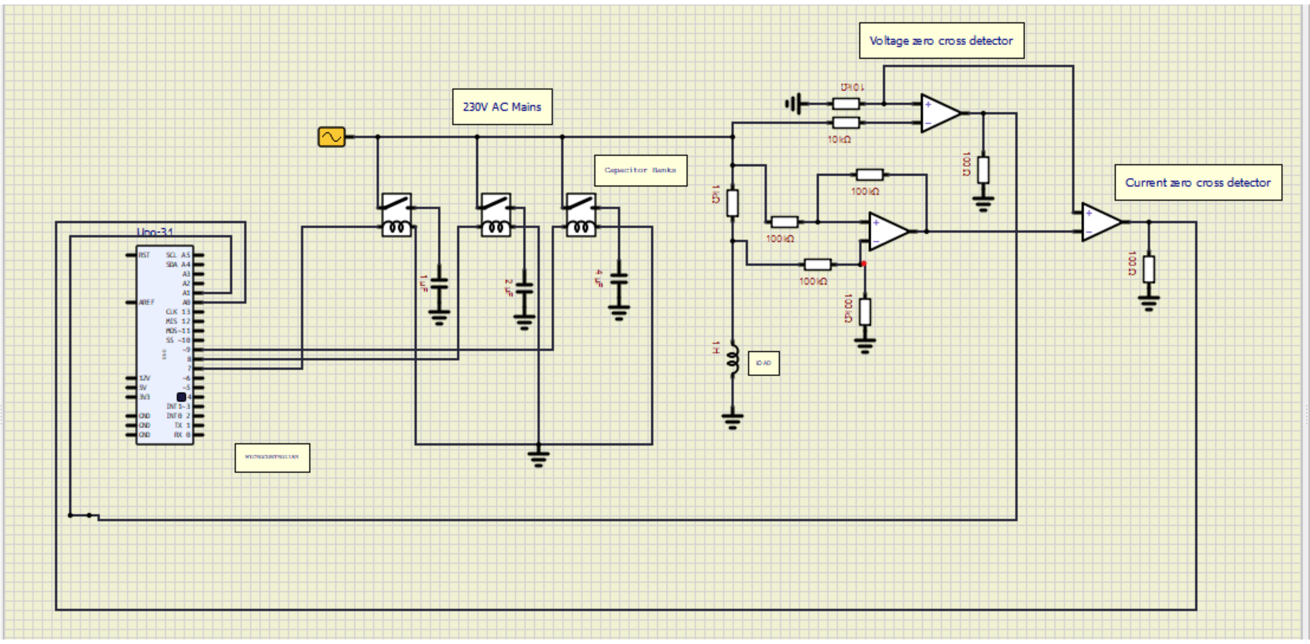
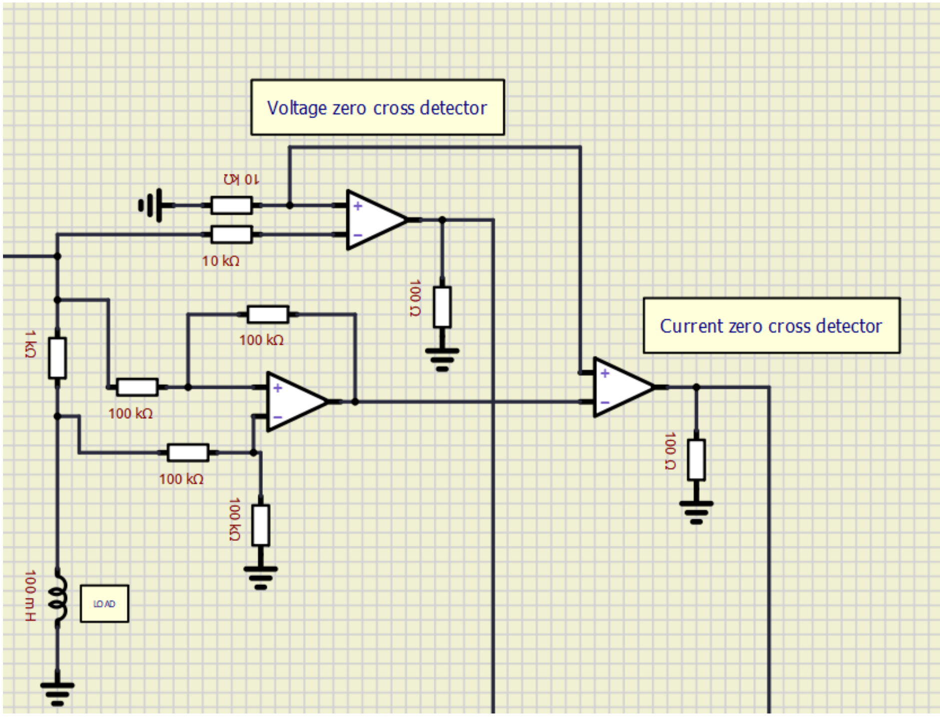
D-03 FINAL TASK

THE FINAL CIRCUIT



The final circuit consists of a microcontroller, capacitor banks (1uF,2uF,4uF) with relay switches for each capacitor , an inductive load and finally the measurement circuit which measure voltage and current through the load.

The Measurement Circuit:



The measurement circuit consists of a Voltage Zero Crossing Detector (VZD) and a Current Zero Crossing Detector (CZD). These are made using opamps.

VCD measures the source voltage and detects when it crosses the zero voltage mark.

The ZCD measures voltage across the resistor (Source voltage – Inductor Voltage) and then detects its zero crossing,

When detected a zero crossing, the circuit generates a pulse. Then the Arduino measures the time delay between the two pulses and then converts to phase delay.

Phase Delay =(time delay/20ms)*360 in degrees

TASK ;

With the VZD & CZD given to Arduino pins A1 & A0 respectively , design a logic which measures the time delay, phase delay then the power factor . Let the threshold power factor be 0.9 . If the power factor goes below 0.9, the the switching sequence should begin

If pf>=0.85 & pf<0.9 then switch on 1uF capacitor

If pf>=0.65 & pf<0.85 then switch on 1uF capacitor & 2uF Capacitor

If pf<0.65 then switch on all the capacitors.

The submissison should consist of a block diagram for the above logic (can be hand drawn) and then should consist of the C++ code for the same . For testing the code u can use a time and phase delay of your choice but make sure phase delay is less than 90 degrees. You can use any compiler or simulation software (TinkerCAD is also ok)

The final code will then be simulated on Simulink MATLAB

Note that your logic will be used for the final simulation hence first understand how Zero crossing Detectors work (Opamps too if u haven't learnt it) t